CHAPTER 19 · REFERENCE DEPTH

ATS Surveillance Phraseology

When a controller can see you on a surveillance display — radar or ADS-B — a whole vocabulary opens up: identifying you, turning you for spacing, warning you of conflicting traffic, ordering an avoiding turn, and managing your transponder. This chapter gathers the surveillance phraseology that appears across approach and area control.

SYLLABUS MAP

Part III (iv) General ATS surveillance — identification & vectoring · traffic info & avoiding action · SSR · radar assistance to comms-failure aircraft · alerting

◎ Learning objectives — by the end of this chapter you will be able to...

- Use identification and vectoring phraseology and respond correctly.
- Receive traffic information and act on avoiding-action instructions.
- Use the full set of SSR (transponder) phraseology.

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- Understand how radar assists an aircraft that has a communications failure.
- Recognise alerting and service-termination phraseology.

19.1 Identification

19.2 Vectoring

19.3 Traffic information & avoiding action

19.4 SSR / transponder phraseology

19.5 Radar assistance to comms failure

19.6 Alerting & termination of service

☆ Numbers to memorise

? Question bank

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When an aircraft is positively identified on a surveillance display, the controller can issue vectors, provide radar separation, and call out conflicting traffic using the clock-code system.

19.1 Identification

FIRST PRINCIPLES — THE CONTROLLER MUST BE SURE WHICH BLIP IS YOU

Before a controller can vector you or give you radar separation, they must **positively identify** your return on the display. They confirm it with "**identified**" (often with a position), or tell you "**not identified**". Common methods are a **squawk ident**, observing a turn, or correlating your reported position.

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Phrase	Meaning
"Squawk [code]"	Set the transponder code
"Squawk ident"	Press IDENT to flash the return for identification
"Identified [position]"	You are identified on radar
"Not identified, resume own navigation"	You are not identified; navigate yourself
"For identification, turn left heading …"	Turn so the controller can identify you

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19.2 Vectoring

HEADINGS FOR A PURPOSE

A **vector** is a heading the controller assigns — for sequencing, spacing, separation or to position for an approach. It usually states the **reason**: "turn left heading two seven zero **for spacing**". When the vectoring is finished, the controller says "**resume own navigation**" (with a position/route).

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TRANSCRIPT — IDENTIFICATION & VECTOR

ATC Ghostair Alfa Bravo Charlie, squawk four three two one and ident.

A/C Squawk four three two one, identing, Ghostair Alfa Bravo Charlie.

ATC Ghostair Alfa Bravo Charlie, identified five miles north of the field, turn right heading one two zero for sequencing.

A/C Right heading one two zero, Ghostair Alfa Bravo Charlie.

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19.3 Traffic information & avoiding action

TRAFFIC INFORMATION

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The controller describes conflicting traffic by its **clock-bearing, distance, direction of movement and (if known) level/type**: "traffic, eleven o'clock, five miles, opposite direction, indicating flight level three five zero". You reply "looking out" and, if you see it, "traffic in sight".

⚠ AVOIDING ACTION — ACT IMMEDIATELY

"**Avoiding action**" is an immediate instruction to turn (and/or climb/descend) to avoid a confliction — for example "**avoiding action, turn left immediately heading one eight zero**, traffic …". It is obeyed at once and read back; the reason follows the instruction so no time is lost.

TRANSCRIPT — TRAFFIC & AVOIDING ACTION

ATC VT-ABC, traffic, two o'clock, four miles, crossing left to right, type unknown.

A/C VT-ABC, looking out … traffic in sight.

ATC VT-ABC, avoiding action, turn left immediately heading three six zero, traffic now one o'clock, two miles.

A/C Avoiding action, left heading three six zero, VT-ABC.

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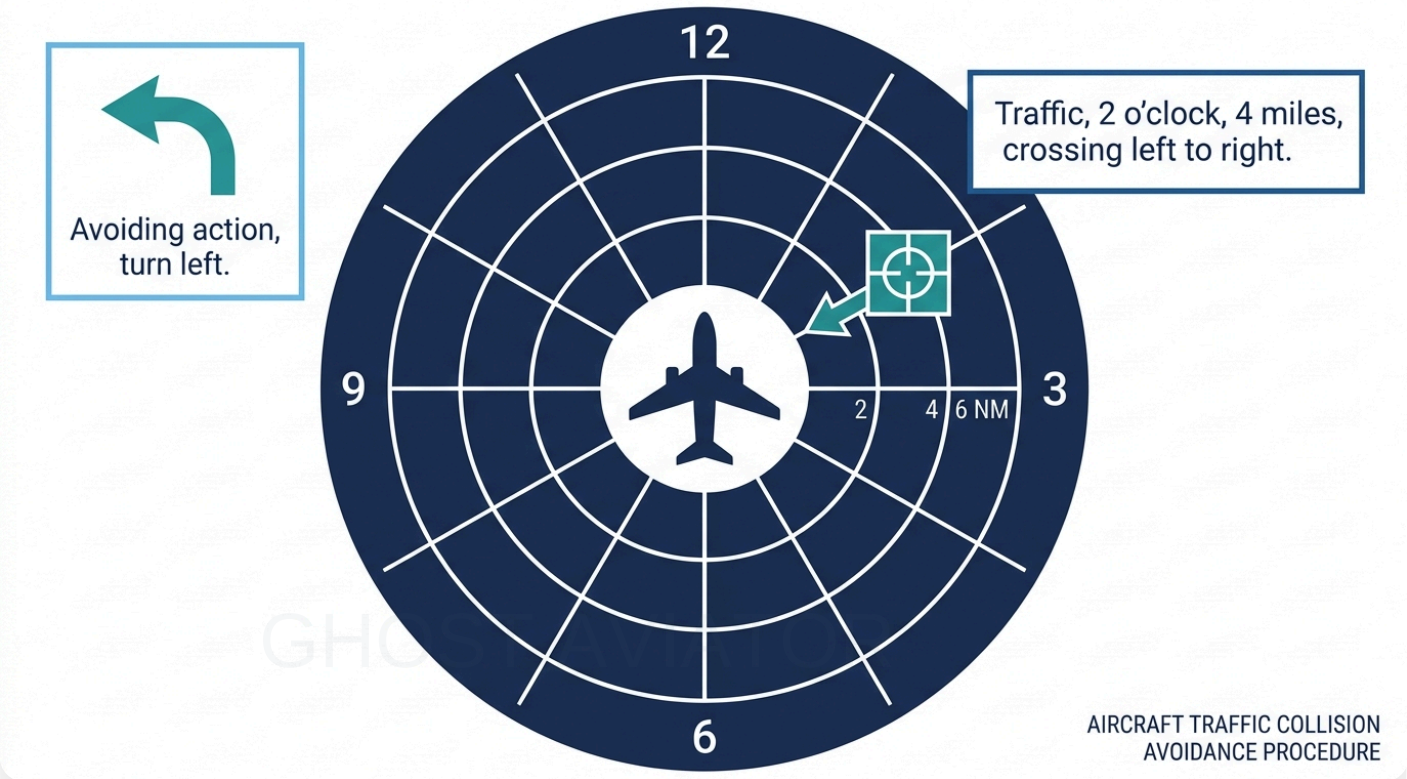


Figure 19.1 — Reading traffic information: The clock code provides a relative bearing to the conflicting traffic based on the aircraft's nose being 12 o'clock.

19.4 SSR / transponder phraseology

Phrase	Meaning
"Squawk [code]"	Set the four-digit code
"Squawk ident"	Operate the IDENT feature

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Phrase	Meaning
"Squawk Charlie" / "squawk Mode C"	Select altitude reporting (Mode C)
"Confirm squawk [code]"	Confirm the code set
"Reset squawk [code]"	Re-select the code (e.g. after a fault)
"Squawk standby"	Select standby (no replies)
"Stop squawk Charlie, wrong indication"	Switch off altitude reporting (faulty)
"Check altimeter setting and confirm level"	Mode C disagrees — verify QNH and level

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SPECIAL CODES (FROM CHAPTER 12)

7500 hijack · **7600** radio failure · **7700** emergency · **7000** conspicuity (VFR) · **2000** entering from a non-SSR area. These are set without being told, in the relevant situation.

19.5 Radar assistance to a comms-failure aircraft

HELPING AN AIRCRAFT THAT CANNOT REPLY

If an aircraft has a **receiver-only** or partial communications failure but its transponder still works, the controller can still **see and assist** it. The controller transmits instructions **blind** and asks the pilot to acknowledge by transponder: "**reply not received, if you read [station], squawk ident**" or "**acknowledge by [a stated transponder action / a turn]**".

⚠ COCKPIT REALITY

Even with no voice, a working transponder turns your aircraft from a guess into a tracked, instructable target — which is why you keep the transponder ON and squawk 7600 the moment a radio fails (full procedure in Chapter 20).

TRANSCRIPT — RADAR ASSISTANCE, RECEIVER STILL WORKING

ATC VT-ABC, if you read Delhi Approach, squawk ident.

ATC VT-ABC, ident observed, I will provide vectors. Turn left heading two seven zero, acknowledge with ident.

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19.6 Alerting & termination of service

POSITION RELATIVE TO KNOWN FEATURES

The controller may give your position relative to a feature for situational awareness or for an aircraft in difficulty: "position five miles east of [VOR]". When radar service ends — leaving cover, or at transfer — the controller says "**identification terminated [instructions]**" or "**radar service terminated, resume own navigation**".

MNEMONIC — THE SURVEILLANCE FLOW

Identify → Vector → Inform/Avoid → (manage SSR) → Terminate. Every surveillance interaction runs through this loop.

☆ Numbers to memorise

Essential Facts for Chapter 19	
Fact	**Value**
Identification	"Identified" / "not identified"; squawk ident, turn, or position correlation

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Fact	Value
Vector	Heading + reason ("for spacing/sequencing"); ends "resume own navigation"
Traffic info	Clock bearing · distance · direction · level/type
Avoiding action	Immediate turn/climb; instruction first, reason after; read back
SSR phrases	Squawk · ident · Charlie (Mode C) · confirm · reset · standby
Comms-failure assistance	Transmit blind; "if you read, squawk ident"; acknowledge by transponder
Termination	"Identification/radar service terminated, resume own navigation"

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Question bank

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Part A — MCQs (click an option to check)

1. "Squawk ident" instructs the pilot to:

Change frequency

Operate the transponder IDENT feature to flash the return

Set 7700

Turn left

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Answer: Operate the transponder IDENT feature to flash the return.

IDENT momentarily highlights the return so the controller can identify it.

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2. "Identified five miles north" means:

You must descend

The controller has positively identified your return

You are not on radar

You must squawk standby

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Answer: The controller has positively identified your return.

It confirms positive radar identification, often with a position.

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3. A vector is:

A squawk code

A heading assigned by the controller, usually with a reason

A frequency

A clearance to land

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Answer: A heading assigned by the controller, usually with a reason.

Vectoring is the assignment of headings for spacing, separation or positioning.

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4. When vectoring ends, the controller says:

"Squawk standby"

"Resume own navigation"

"Cleared to land"

"Go around"

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Answer: "Resume own navigation".

"Resume own navigation" returns navigation responsibility to the pilot.

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5. Traffic information is given as:

A frequency and a level

Clock bearing, distance, direction of movement, and level/type if known

A squawk code

A QNH

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Answer: Clock bearing, distance, direction of movement, and level/type if known.

e.g. "traffic, two o'clock, four miles, crossing left to right".

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6. "Avoiding action, turn left immediately heading 180" must be:

Considered and queried first

Acted on at once and read back

Ignored if VFR

Acknowledged with "roger"

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Answer: Acted on at once and read back.

Avoiding action is an immediate manoeuvre, executed at once and read back.

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7. "Squawk Charlie" instructs the pilot to:

Set code 7700

Select Mode C (altitude reporting)

Squawk standby

Ident

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Answer: Select Mode C (altitude reporting).

"Charlie"/Mode C turns on pressure-altitude reporting.

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8. "Squawk standby" means:

Reply normally

Select standby — the transponder does not reply

Set 2000

Ident

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Answer: Select standby — the transponder does not reply.

Standby stops the transponder replying (e.g. on the ground).

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9. If a controller suspects the Mode C readout is wrong, they may say:

"Squawk ident"

"Check altimeter setting and confirm level" / "stop squawk Charlie, wrong indication"

"Resume own navigation"

"Cleared to land"

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Answer: "Check altimeter setting and confirm level" / "stop squawk Charlie, wrong indication".

A Mode C disagreement prompts a level/altimeter check or stopping Mode C.

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10. A controller can still assist an aircraft with a receiver-only failure if it has:

No transponder

A working transponder (so it can be tracked and instructed blind)

A working ELT only

Only HF

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Answer: A working transponder (so it can be tracked and instructed blind).

A working transponder lets the controller track it and pass blind instructions acknowledged by squawk.

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11. To get a no-voice aircraft to confirm it can hear, the controller says:

"Confirm visual"

"If you read [station], squawk ident"

"Squawk standby"

"Resume own navigation"

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Answer: "If you read [station], squawk ident".

An ident in response confirms the pilot is receiving.

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12. "Resume own navigation" returns to the pilot:

The landing clearance

Responsibility for navigating the aircraft

The squawk code

The frequency

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Answer: Responsibility for navigating the aircraft.

It ends ATC heading-vectoring and the pilot navigates again.

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13. The transponder code for general emergency is:

7500

7600

7700

7000

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Answer: 7700.

7700 emergency, 7600 radio failure, 7500 hijack.

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14. "Traffic in sight" tells the controller:

You will not avoid it

You have visually acquired the reported traffic

You have a radio failure

You are identified

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Answer: You have visually acquired the reported traffic.

It confirms you can see the traffic the controller described.

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15. Radar service termination is phrased as:

"Squawk ident"

"Radar service terminated, resume own navigation"

"Cleared to land"

"Avoiding action"

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Answer: "Radar service terminated, resume own navigation".

When leaving cover or at transfer, the radar/identification service is terminated.

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16. The general surveillance interaction sequence is:

Land → taxi → park

Identify → vector → inform/avoid → manage SSR → terminate

Climb → cruise → descend

Squawk → ident → land

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Answer: Identify → vector → inform/avoid → manage SSR → terminate.

Identify, vector, give traffic info/avoiding action, manage the transponder, then terminate.

Part B — Oral / viva (tap to reveal model answers)

▼ ? **How does a controller identify an aircraft on radar?**

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Model Answer:

By a transponder ident ("squawk ident"), by observing an instructed turn, or by correlating the aircraft's reported position; the controller then confirms "identified", usually with a position.

▼ ? **What is a vector, and how is the service ended?**

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Model Answer:

A vector is a heading assigned by the controller, usually with a reason such as spacing or sequencing. When complete the controller says "resume own navigation".

▼ ? How is traffic information given, and what is avoiding action?

Model Answer:

Traffic is described by clock bearing, distance, direction of movement and, if known, level or type. Avoiding action is an immediate instruction to turn or change level to avoid a confliction, given instruction-first then reason, executed at once and read back.

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▼ ? Give five SSR phraseology items and their meanings.

Model Answer:

"Squawk [code]" — set the code; "squawk ident" — operate IDENT; "squawk Charlie/Mode C" — select altitude reporting; "confirm/reset squawk" — verify/re-select; "squawk standby" — select standby so it does not reply.

▼ ? How can a controller assist an aircraft with a communications failure?

Model Answer:

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If the aircraft can still receive and its transponder works, the controller transmits instructions blind and asks the pilot to acknowledge by transponder — "if you read, squawk ident" — and then provides vectors, with each acknowledged by ident.

⚡ 60-SECOND REVISION CARD

Identify (squawk ident / turn / position) → "identified" or "not identified".

Vector = heading + reason; ends "**resume own navigation**".

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Traffic = **clock · distance · direction · level/type**. **Avoiding action** = immediate, read back.

SSR: squawk · ident · **Charlie (Mode C)** · confirm · reset · standby. Codes **7500/7600/7700/7000/2000**.

Comms failure: "**if you read, squawk ident**", vectors acknowledged by transponder. Service ends "terminated, resume own navigation".

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